

Daily Integral - Medium

$$\int_0^{\pi/4} \frac{\ln(1 - \cos(2x)) - \ln(1 + \cos(2x))}{1 + \cos(2x) + \sin(2x)} dx$$

This problem was generously submitted by Archie B. on <https://dailyintegral.com>.

Solution

First, combine the natural logs and use the half-angle identities and the tangent quotient identities:

$$\begin{aligned} & \int_0^{\pi/4} \frac{\ln\left(\frac{1 - \cos(2x)}{1 + \cos(2x)}\right)}{1 + \cos(2x) + \sin(2x)} dx \\ &= \int_0^{\pi/4} \frac{\ln\left(\frac{\sin(x)^2}{\cos(x)^2}\right)}{1 + \cos(2x) + \sin(2x)} dx \\ &= \int_0^{\pi/4} \frac{\ln(\tan(x)^2)}{1 + \cos(2x) + \sin(2x)} dx \\ &= \int_0^{\pi/4} \frac{2 \ln(\tan(x))}{1 + \cos(2x) + \sin(2x)} dx. \end{aligned}$$

Now, use the Weierstrass substitution and take $t = \tan(x)$. Then, $\sin(2x) = 2t / (1 + t^2)$, $\cos(2x) = (1 - t^2) / (1 + t^2)$, and $dx = 1 / (1 + t^2) dt$, with $t \in [0, 1]$:

$$\begin{aligned} & \int_0^1 \frac{2 \ln(t)}{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \frac{1}{1+t^2} dt \\ &= \int_0^1 \frac{2 \ln(t)}{1 + t^2 + 1 - t^2 + 2t} dt \\ &= \int_0^1 \frac{\ln(t)}{1+t} dt. \end{aligned}$$

Now, use the Maclaurin series $1/(1+t) = \sum_{n=0}^{\infty} (-1)^n t^n$:

$$\begin{aligned} & \int_0^1 \ln(t) \sum_{n=0}^{\infty} (-1)^n t^n dt \\ &= \sum_{n=0}^{\infty} (-1)^n \int_0^1 \ln(t) t^n dt. \end{aligned}$$

To perform integration by parts, take $u = \ln(t)$ and $dv = t^n$. Then, $du = 1/t dt$, and $v = t^{n+1}/(n+1)$:

$$\begin{aligned} & \sum_{n=0}^{\infty} (-1)^n \left(\frac{\ln(t) t^{n+1}}{n+1} - \int \frac{t^n}{n+1} dt \Big|_0^1 \right) \\ &= \sum_{n=0}^{\infty} (-1)^n \left(\frac{\ln(t) t^{n+1}}{n+1} - \frac{t^{n+1}}{(n+1)^2} \Big|_0^1 \right) \\ &= \sum_{n=0}^{\infty} (-1)^n \left(\frac{-1}{(n+1)^2} \right) \\ &= \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(n+1)^2} \\ &= -1 + \frac{1}{4} - \frac{1}{9} + \frac{1}{16} - \frac{1}{25} - \dots \\ &\approx -0.82. \end{aligned}$$